

Project Title:

Hasta-Shilpa – Design Bridge for Artisans

Student Details:

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 - **Project Number:** 32
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1. Problem Statement

Bamboo and cane artisans in rural regions create high-quality handcrafted products. However, their designs are often limited to traditional patterns, which do not align with modern urban market trends. This results in low demand and reduced income. There is a need for a solution that introduces modern design ideas while preserving traditional craftsmanship.

2. Proposed Solution

The proposed application, *Hasta-Shilpa*, acts as a **design bridge** between traditional artisans and modern market needs. The app provides a curated design library, showcasing trending bamboo and cane products along with detailed measurement guides (blueprints). It enables artisans to replicate modern designs easily and price their products appropriately.

3. Objectives

- To provide artisans with access to **modern design inspirations**
- To help artisans create products with **accurate measurements**
- To assist in **pricing products** based on materials and effort
- To promote **sustainable materials** like bamboo
- To improve the **market value of handmade products**

4. Key Features

- **Design Trend Feed:** Displays modern bamboo/cane product ideas
 - **Blueprint View:** Provides sketches with dimensions for easy replication
 - **Material Tracker:** Allows artisans to log materials used
 - **Price Suggester:** Calculates product price based on inputs
 - **Marketplace (Simulated):** Displays finished products for listing
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5. User Flow

1. User opens the app and browses trending designs
 2. Selects a design to view blueprint and measurements
 3. Uses the material tracker to record resources used
 4. Inputs cost and working hours into the price suggester
 5. Views estimated product price and lists the product
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6. Technical Implementation

- **Platform:** Android Studio
 - **Language:** Kotlin
 - **UI Framework:** Jetpack Compose
 - **Storage:** Local storage (Room Database / simple data handling)
 - **UI Components:** Cards, Image Viewer, Input Forms
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7. Expected Outcomes

- Increased creativity among artisans
 - Improved product quality and market relevance
 - Higher income through better pricing strategies
 - Promotion of eco-friendly materials
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8. Feasibility Analysis

The project is feasible as it primarily involves UI design and basic logic implementation. It does not require complex integrations like GPS or real-time databases, making it suitable for development within the given timeframe.

9. Future Enhancements

- Integration of AI-based design suggestions
- Real-time marketplace for selling products
- Multi-language support for rural users
- Cloud storage for design sharing

10. Conclusion

Hasta-Shilpa is a practical and impactful solution that bridges the gap between traditional craftsmanship and modern design demands. It empowers artisans with the tools needed to innovate, improve their income, and sustain their craft in a competitive market.